DELHI PUBLIC SCHOOL JANKIPURAM

SET–II

HALF YEARLY EXAMINATION: (2024-25)

SUBJECT: MATHEMATICS (041)

GRADE: IX

Time – 3 hrs. **M.M. 80**

Name- _____ **Roll No-** _____ **Date-** _____

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General Instructions:

Read the following instructions carefully and follow them:

- *All questions are compulsory*
- *Section A: Question 1-18 are objective type and conceptual based and question 19 and 20 are assertion and reason type each consist of 1 mark each.*
- *Section B: Question 21-25 consist of 2 marks each.*
- *Section C: Question 26-31 consist of 3 marks each.*
- *Section D: Question 32-35 consist of 5 marks each.*
- *Section E: Question 36-38 consist of 4 marks each containing two case study based questions with subparts of 1+1+2 marks.*
- *Internal choices are provided in each section. Use of calculator is not permitted.*
- *This question paper consists of 6 pages.*

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SECTION A

Directions: Question no. 1 to 18 are MCQ / Conceptual approach and question 19 and 20 are assertion and reason type each consist of 1 mark.

1. The exponent form of $\sqrt[3]{7}$ is: 1

(a) 7^3 (b) 3^7 (c) $7^{1/3}$ (d) $3^{1/7}$

2. The equation $x = 7$, in two variables, can be written as 1

(a) $x + 0y = 7$ (b) $0x + y = 7$ (c) $0x + 0y = 7$ (d) $x + y = 7$

3. On rationalizing the denominator of $\frac{1}{2+\sqrt{3}}$, we get 1

(a) $2-\sqrt{3}$ (b) $\sqrt{3}-2$ (c) $2+\sqrt{3}$ (d) $-\sqrt{3}-2$

4. Zero of the polynomial $p(x) = a^2x$, $a \neq 0$ is 1

(a) $x = 0$ (b) $x = 1$ (c) $x = -1$ (d) $a = 0$

5. If the coordinates of two points P and Q are (2, –3) and (–6, 5), then the value of (x-coordinate of P) – (x-coordinate of Q) is 1

(a) 2 (b) –6 (c) –8 (d) 8

6. Class mark of class 150 – 160 is 1

(a) 150 (b) 160 (c) 155 (d) none of these.

7. Which of the following is true? 1

(a) Every whole number is a natural number (b) Every integer is a rational number

(c) Every rational number is an integer (d) Every integer is a whole number

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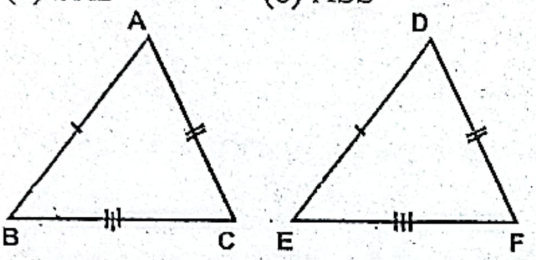
8. The origin lies on
(a) x-axis only (b) both axes (c) y-axis only (d) none of the axes

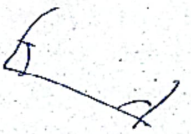
9. What is the common between the three angles of a triangle and a linear pair? 1
(a) angles are equal (b) in both cases sum of angle is 180^0.
(c) In triangle there are three angles and in linear pair there are two angles (d) none of these.

10. Out of the following, the irrational number is 1
(a) $1.\bar{5}$ (b) $2.4\overline{77}$ (c) $1.2\overline{77}$ (d) π

11. If an angle of a parallelogram is two-third of its adjacent angle, the smallest angle of the parallelogram is 1
(a) 108^0 (b) 54^0 (c) 72^0 (d) 81^0

12. By which congruence rule following triangles are congruent? 1
(a) SAS (b) ASS (c) AAA (d) SSS





13. The point (–5, 4) and (4, –5) are situated in 1
(a) same quadrant (b) I and III quadrant, respectively
(c) Different quadrants (d) IV and II quadrant, respectively

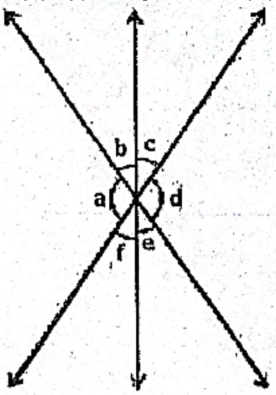
14. The angle which is two times its complement is 1
(a) 60^0 (b) 30^0 c) 45^0 (d) 72^0

15. ABCD is a rhombus such that $\angle ACB = 40^0$, then $\angle ADB =$ 1
(a) 45^0 (b) 50^0 (c) 40^0 (d) 60^0

16. The line segment joining the midpoints of two sides of a triangle is parallel to the third side and _____ of it. 1
(a) half (b) one third (c) one fourth (d) equal

17. Line segment joining the mid point of any side with the opposite vertex is 1
(a) altitude (b) median c) perpendicular bisector (d) angle bisector

18. In figure, which of the following statements must be true? 1
(i) $a + b = d + c$ (ii) $a + c + e = 180^0$ (iii) $b + f = c + e$
(a) (i) only (b) (ii) only c) (iii) only (d) (ii) and (iii) both



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19. **Assertion:** 0.468 is a terminating decimal. 1
Reason: a decimal in which a digit or a set of digits is repeated periodically, is called a repeating, or a recurring decimal.
a) both Assertion and Reason are correct and Reason is the correct explanation for Assertion
b) both Assertion and Reason are correct and Reason is not the correct explanation for Assertion.
c) assertion is true but the reason is false.
d) both assertion and reason are false.

20. **Assertion:** a line segment is a part of a line with two end-points. 1
Reason: a line can be extended from both side
a) both Assertion and Reason are correct and Reason is the correct explanation for Assertion
b) both Assertion and Reason are correct and Reason is not the correct explanation for Assertion.
c) Assertion is true but the Reason is false.
d) both Assertion and Reason are false.

SECTION B

Directions: *Questions no. Q21 to Q25 are Short answer type questions of 2 marks each.*

21. Simplify: 2

$$\left[5\left(8^{\frac{1}{3}}+27^{\frac{1}{3}}\right)^{3}\right]^{\frac{1}{4}}$$

22. If the point (3, 4) lies on the graph of $3y = ax + 7$, then find the value of a. 2

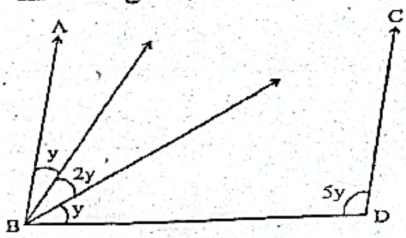
OR

Determine the point on the graph of the equation $2x + 5y = 20$ whose x-coordinate is $\frac{5}{2}$ times its ordinate.

23. Find the value of k, if $x - 1$ is a factor of $4x^3 + 3x^2 - 4x + k$. 2

24. Prove that opposite angles of a parallelogram are equal. 2

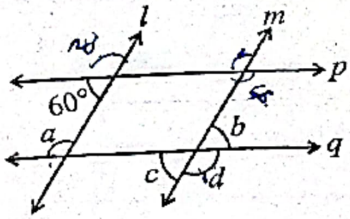
25. In the figure, if AB || CD, then what is the value of y. 2



OR



Given Lines $l \parallel m$, $p \parallel q$; Find a, b, c, d

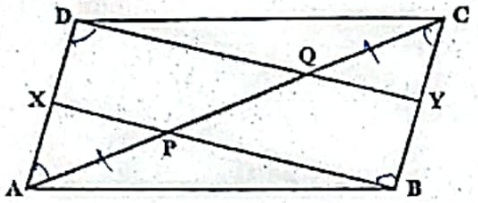


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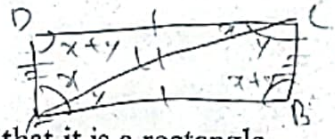
SECTION C

Directions: Questions no. Q26 to Q 31 are Short answer type questions of 3 marks each.

26. In the below figure, X and Y are respectively the mid-points of the opposite sides AD and BC of a parallelogram ABCD. Also, BX and DY intersect AC at P and Q, respectively. Show that AP = PQ = QC. 3

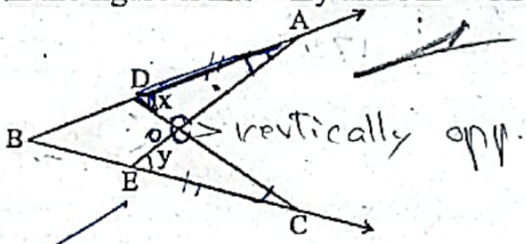


OR



If the diagonals of a parallelogram are equal, then show that it is a rectangle.

27. In the figure if $\angle x = \angle y$ and AB = CB. Prove that AE = CD. 3



28. A transversal intersects two parallel lines. Prove that the bisectors of any pair of corresponding angles so formed are parallel. 3

29. A three - wheeler scooter charges Rs.10 for first kilometer and Rs.4.50 each for every subsequent kilometer. For a distance of x km, an amount of Rs.y is paid. Write the linear equation representing the above information. Also find the amount to be paid if the distance travelled is 15km. 3

30. If $a + b + c = 5$ and $ab + bc + ca = 10$, then prove that $a^3 + b^3 + c^3 - 3abc = -25$. 3

OR

If both $x - 2$ and $x - \frac{1}{2}$ are factors of $px^2 + 5x + r$, show that $p = r$.

31. Find the coordinates of the point 3

(i) which lies on x and y axes both.

(ii) whose ordinate is -4 and which lies on y-axis

(iii) whose abscissa is 5 and which lies on x-axis.

SECTION D

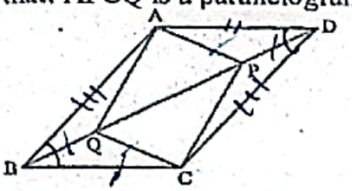
Directions: Question 32-35 are long answer type questions of 5 marks each

32. Draw a histogram and the frequency polygon from the following data. 5

Class interval	20–25	25–30	30–35	35–40	40–45	45–50
Frequency	30	24	52	28	46	10

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33. In parallelogram ABCD, two points P and Q are taken on diagonal BD such that DP = BQ. Show that: APCQ is a parallelogram 5



5

34. Represent the real number $\sqrt{13}$ on the number line. 5

5

OR

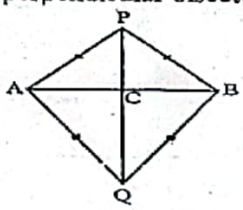
Simplify $\frac{4+\sqrt{5}}{4-\sqrt{5}}+\frac{4-\sqrt{5}}{4+\sqrt{5}}$ by rationalizing the denominator.

35. In an isosceles triangle ABC, with AB = AC, the bisectors of ∠B and ∠C intersect each other at O. Join A to O. Show that : (i) OB = OC (ii) AO bisects ∠A 5

1½

OR

AB is a line-segment. P and Q are points on opposite sides of AB such that each of them is equidistant from the points A and B(see in the below left figure). Show that the line PQ is the perpendicular bisector of AB.

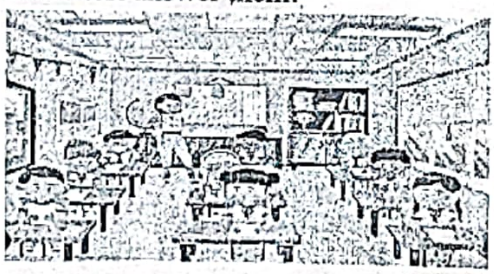


SECTION E

Directions: Question 36-38 are case study based questions of 4-marks with subparts of 1+1+2 marks.

36. **CASE STUDY - I:** 4

Group study offers several key benefits. It enhances understanding by providing diverse perspectives and collaborative problem-solving. Students improve their communication skills by articulating ideas and actively listening to peers.. Group study also helps develop teamwork skills, effective time management, and provides emotional support.. A teacher decided to conduct a group quiz test related to polynomials after the completion of the chapter. Following questions were asked in the test answer them:



$p+q=17$
$p\times q=30$

2

1	30
32	15
5	5
	1

(a) What is the degree of constant polynomial? 1
(b) Find the zero of the polynomial $p(x) = 2x + 1$ −0.5 1
(c) Find the value of a, if x − a is a factor of $x^3 - ax^2 + 2x + a - 1$. 2

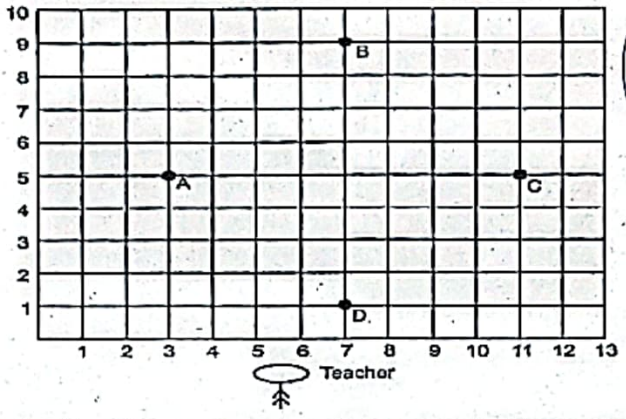
OR

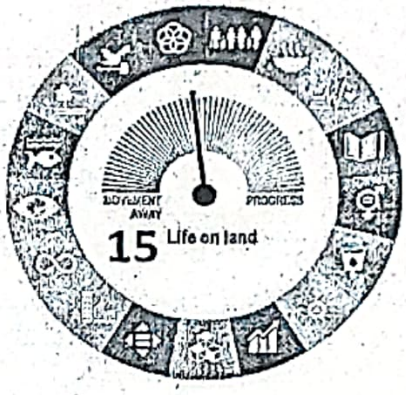
Factorise $6x^2 + 17x + 5$

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37. **CASE STUDY - II: SDG 15: Life on Land** 4

Students of a school are standing in rows and columns on their playground to create the perfect shape for gardening plantation, **contributing to SDG 15: Life on Land.** This activity promotes sustainable land use and encourages environmental stewardship by engaging students in hands-on gardening, which helps improve green spaces and fosters a connection to nature. A, B, C and D are the positions of four students as shown in the figure.





(a) What are the coordinates of the point of intersection of AC and BD? (7, 5) 1

(b) What is the distance between B and D? 8 units 1

(c) What are the coordinates of A and B respectively? (3, 5) (7, 9) 2

OR

If the teacher is taking round and stands at the point of intersection of both the axes then write its coordinate, what it is the name of this point?

38. **CASE STUDY - III:** 4

Statistics is the study of the collection, analysis, interpretation, presentation, and organization of data. In other words, it is a mathematical discipline to collect, summarize data. A group of students decided to make a project on Statistics. They are collecting the heights (in cm) of their 51 girls of Class IX-A, B and C of their school. After collecting the data, they arranged the data in the following frequency distribution table form:shown in adjoining figure.

Height (in cm)	Number of girls
135 – 140	4
140 – 145	7
145 – 150	18
150 – 155	11
155 – 160	6
160 – 165	5



(a) How many students of the height more than 145 cm but less than 155 cm are there? 29 1

(b) What is the width of the class? 5 1

(c) Find the class mark of the class interval having highest frequency. 147.5 2

OR

Find the difference between the upper limit of class interval having highest frequency and the lower limit of the class interval having lowest frequency.

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